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Provisioning AWS services ...
Estimating the time remaining. Most labs take about 5 minutes, but it could take longer

Building Data Lakes on AWS - Automate Data Lake Creation using AWS Lake Formation Blueprints

SPL-CX-200-ANABDL-1 – Version 1.0.1

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Note: Do not include any personal, identifying, or confidential information into the lab environment. Information entered may be visible to others.

Corrections, feedback, or other questions? Contact us at AWS Training and Certification.

Lab overview

You are a data engineer at AnyCompany, a cloud marketing organization. You are asked to populate a data lake with content from AWS CloudTrail so the operational analytics team can efficiently query their data with Amazon Athena.

In this lab, you use a workflow provided as an AWS Lake Formation blueprint to greatly simplify the creation of a data lake and ingestion of data. Lake Formation blueprints are workflows you can apply to an existing Lake Formation data lake. You can also apply them as a task in the setup and creation of a new data lake.

Objectives

By the end of this lab, you will be able to do the following:

- Create an AWS Glue workflow using a Lake Formation blueprint.
- Automate the Lake Formation data lake setup process with an AWS Glue workflow.
- Create a custom AWS Glue workflow.

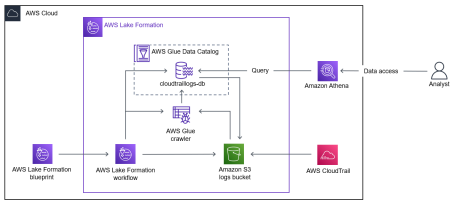
Icon key

Various icons are used throughout this lab to call attention to different types of instructions and notes. The following list explains the purpose for each icon:

- **Note:** A hint, tip, or important guidance.
- **Consider:** A moment to pause to consider how you might apply a concept in your own environment or to initiate a conversation about the topic at hand.
- **Hint:** A hint to a question or challenge.
- **Answer:** An answer to a question or challenge.

Environment overview

The following diagram shows the basic architecture of the lab environment:



In the preceding diagram, a data lake is created using a Lake Formation blueprint. The blueprint creates a workflow that uses an Amazon Simple Storage Service (Amazon S3) bucket connected to CloudTrail logs, a crawler, and a database. The workflow loads the CloudTrail log data stored in Amazon S3 into the database in the AWS Glue Data Catalog. An analyst queries the Lake Formation database with Athena. Athena uses the AWS Glue Data Catalog and reads data from the database.

Start lab

1. To launch the lab, at the top of the page, choose **Start Lab**.
 - **Caution:** You must wait for the provisioned AWS services to be ready before you can continue.
2. To open the lab, choose **Open Console**.
 - You are automatically signed in to the AWS Management Console in a new web browser tab.
 - **Warning:** Do not change the **Region** unless instructed.

Common sign-in errors

Error: You must first sign out



If you see the message, **You must first log out before logging into a different AWS account:**

- Choose the **click here** link.
- Close your **Amazon Web Services Sign In** web browser tab and return to your initial lab page.
- Choose **Open Console** again.

Error: Choosing Start Lab has no effect

In some cases, certain pop-up or script blocker web browser extensions might prevent the **Start Lab** button from working as intended. If you experience an issue starting the lab:

- Add the lab domain name to your pop-up or script blocker's allow list or turn it off.
- Refresh the page and try again.

Task 1: Explore the lab environment

In this task, you review the account resources created before the lab started.

1. At the top of the AWS Management Console, in the search bar, search for and choose `S3`.
4. Choose the link for the bucket name that starts with **cloudtraildatabucket**.

Note: The **data/** folder will contain your dataset. The **results/** folder stores the results of your Athena queries. You will specify the **results/** folder as the query results location in Athena later in this lab.

5. Choose the **data/** folder. You will specify the **data/** folder as your designated storage location for your data lake later in this lab.
6. At the top of the AWS Management Console, in the search bar, search for and choose `CloudTrail`.
7. In the **Trails** section, choose **LabCloudTrail**.

Note: If you receive an error related to "Access Denied" or "You do not have permission to perform this action" on cloudtrail console in rest of lab, then please ignore and proceed through next steps.

Consider: Which bucket are the logs stored in?

You use this trail as part of your Lake Formation blueprint.

Congratulations! You successfully explored the lab environment.

Task 2: Set up Lake Formation

In this task, you register your Amazon S3 data storage and create a database.

Task 2.1: Register your Amazon S3 storage

Lake Formation manages access to designated storage locations in Amazon S3. Register the storage locations you want to be part of the data lake.

8. At the top of the AWS Management Console, in the search bar, search for and choose `AWS Lake Formation`.
9. Close any popups
10. Go to Administrative roles and tasks under **Administration** and choose **Add** under **Data lake administrators**.
11. Under **IAM users and roles** find and click on your federated user name-role (starts with AWSLabsUser-<random-number>) and click on **Confirm**
12. In the left navigation pane, in the **Administration** section, choose **Data lake locations**.
13. Choose **Register location**.
14. On the **Register location** page, in the **Amazon S3 location** section:
 - For **Amazon S3 path**, copy and paste the **SourceDataLocation** value that is listed to the left of these instructions.
 - For **IAM role**, select **LakeFormationServiceRole**.
 - For **Permission mode**, select **Lake Formation**.
15. Choose **Register location**.
16. In the left navigation pane, in the **Permissions** section, choose **Data locations**.
17. Choose **Grant**.
18. On the **Grant permissions** page, configure the following:
 - For **IAM users and roles**, select **LakeFormationWorkflowRole**.
 - For **Storage locations**, copy and paste the **SourceDataLocation** value that is listed to the left of these instructions.
19. Choose **Grant**.

Your Amazon S3 bucket **data/** folder is now registered as the storage location for your data lake.

Task 2.2: Create a database

Lake Formation organizes data into a catalog of logical databases and tables. Create a database in the AWS Glue Data Catalog.

20. In the left navigation pane, in the **Data catalog** section, choose **Databases**.
21. Choose **Create database**.
22. On the **Create database** page, in the **Database details** section:
 - For **Name**, enter `cloudtrailogs-db`.
 - For **Location – optional**, copy and paste the **SourceDataLocation** value that is listed to the left of these instructions.
23. Choose **Create database**.

You created a database in the AWS Glue Data Catalog using Lake Formation.

Congratulations! You successfully registered your Amazon S3 data storage and created a database.

Task 3: Use a Lake Formation blueprint to create an AWS Glue workflow

In this task, you use a Lake Formation blueprint to create an AWS Glue workflow that will automatically add new content to your data lake.

A workflow encapsulates a complex multi-job extract, transform, and load (ETL) activity. Workflows generate AWS Glue crawlers, jobs, and triggers to orchestrate the loading and updating of data. Lake Formation runs and tracks a workflow as a single entity. You can configure a workflow to run on-demand or on a schedule.

Workflows you create in Lake Formation are visible in the AWS Glue console as a directed acyclic graph (DAG). Each DAG node is a job, crawler, or trigger. To monitor progress and troubleshoot, you can track the status of each node in the workflow.

24. In the left navigation pane, in the **Ingestion** section, choose **Blueprints**.
25. Choose **Use blueprint**.
26. On the **Use a blueprint** page:
 - For **Blueprint type**, choose **AWS CloudTrail**.
 - For **CloudTrail name**, choose **LabCloudTrail**.
 - For **Start date**, choose today's date.
 - For **Target database**, choose **cloudtrailogs-db**.
 - For **Target storage location**, add the **S3 bucket** from the left of instructions.
 - For **Data format**, choose **Parquet**.
 - For **Frequency**, choose **Run on demand**.
 - For **Workflow name**, enter `lf-cloudtrail-workflow`.
 - For **IAM role**, choose **LakeFormationWorkflowRole**.
 - For **Table prefix**, enter `lab`.
27. Choose **Create**.

Choose the refresh icon in the **Workflows** section until you see your new workflow.

Congratulations! You successfully used a Lake Formation blueprint to create an AWS Glue workflow that will automatically add new content to your data lake.

Task 4: Run and monitor the workflow

In this task, you run an AWS Glue workflow. While it is running, explore AWS Glue workflows and monitor your workflow until it is complete.

Task 4.1: Run the workflow

Run a workflow and view the DAG in AWS Glue.

28. Choose the link for **lf-cloudtrail-workflow**.
29. Choose **Start**.
30. Refresh your browser tab.

In the **Workflow runs** section, a new run appears.

31. At the top of the AWS Management Console, in the search bar, search for and choose `AWS Glue`.
32. In the left navigation pane, in the **Data integration and ETL** section, choose **Workflows (orchestration)**.

Note: If the **Workflows** page does not load, choose **Workflows** again.

The **Last run status** for the **lf-cloudtrail-workflow** is **Running**.

33. Choose **lf-cloudtrail-workflow**.
34. Choose the **History** tab.
35. Choose the run that starts with **wr_**.
36. Choose **View run details**.

The workflow DAG appears.

The workflow takes 15–20 minutes to run. You can view the workflow progress using the DAG in the **Graph** section.

Consider: Take a moment to view the DAG. Which nodes have already completed?

While you wait for this workflow to complete, continue with the next task.

Task 4.2: Explore AWS Glue workflows

While your workflow is running, build a sample AWS Glue workflow and explore the DAG.

37. In the left navigation pane, in the **Data integration and ETL** section, choose **Workflows (orchestration)**.
38. Choose **Add workflow**.
39. For **Workflow name**, enter `sample-workflow`.
40. Choose **Create workflow**.
41. Choose **sample-workflow**.


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